

Math 252 - Calculus II - Midterm Exam 1

CRN: 23526

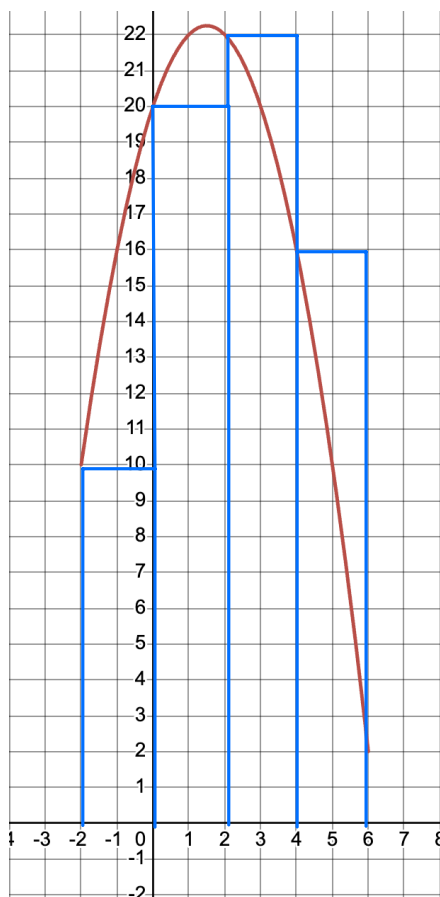
January 31, 2025

Last Name: _____, First Name: _____

1. There are 6 problems. Many problems have multiple parts. The total credit is 100.
2. No collaboration is allowed for this exam. The work on this exam is to be yours and yours alone. Failure to adhere to this policy will result in a zero on this exam. Any instances of suspected cheating will also be reported to the Dean of Students' Office.
3. No calculators, books or other material are permitted.
4. In order to receive full credit you need to show your work, use complete sentences to explain your answers, and present your answers as clearly as possible.
5. Answers must be exact (like $\sqrt{2}$ or π), not approximate (like 1.414, 3.14), unless the problem specifically indicates otherwise.
6. Simplify where appropriate. Quantities like $\sqrt{9}$, $\ln e$, inverse trigonometric functions at special values, or trigonometric functions at special angles should be calculated. Reduce all fractions.
7. If an answer line is provided for the answer, place your final answer in it. Otherwise, circle, box, or underline each final answer as appropriate for clarity.
8. Failure to follow directions specific to a problem will result in loss of points.
9. Put away and **silence** your cell phones, tablets, laptops, etc.
10. Once this exam begins, you will have 50 minutes to complete your solutions.

1. [10 pts] Consider the function $f(x) = -x^2 + 3x + 20$ on the interval $[-2, 6]$. The graph of the function is given below.

- 1). Illustrate the left-endpoint Riemann sum L_4 of $f(x)$ on the interval $[-2, 6]$ with $n = 4$ as area of rectangles by sketching the appropriate rectangles on the graph.
- 2). Calculate the left-endpoint Riemann sum L_4 of $f(x)$ on the interval $[-2, 6]$ with $n = 4$. Simplify your answer.



$$\Delta x = 2$$

Left-end points: $-2, 0, 2, 4$

Heights: $f(-2) = -4 - 6 + 20 = 10$

$$f(0) = 20$$

$$f(2) = -4 + 6 + 20 = 22$$

$$f(4) = -16 + 12 + 20 = 16$$

$$L_4 = \text{Area} = 10 \cdot 2 + 20 \cdot 2 + 22 \cdot 2 + 16 \cdot 2$$

$$= 20 + 40 + 44 + 32$$

$$= 136$$

2. [7 pts] Consider the function $f(x) = x \tan(2 + x)$ on $[1, 4]$.

- a). If the interval $[1, 4]$ is divided into n subintervals of equal length, what is the width of each subinterval?

Answer: The width of each subinterval $\Delta x = \underline{\frac{3}{n}}$

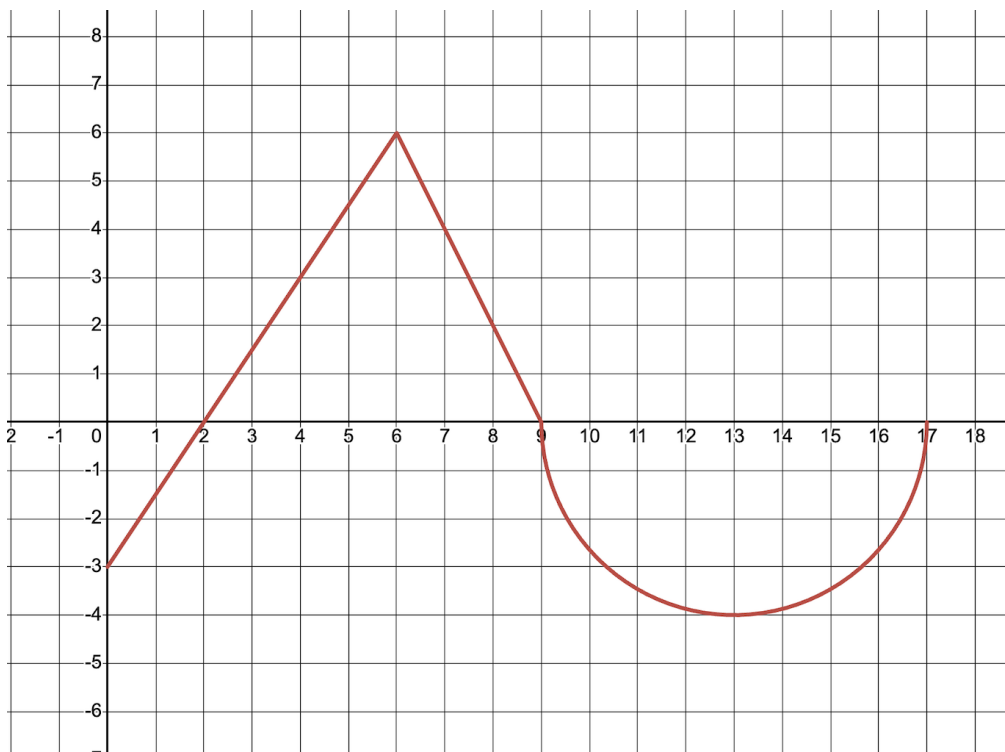
- b). Find the right-endpoint x_i of the i th subinterval, and then find $f(x_i)$.

Answer: $x_i = \underline{1 + \frac{3i}{n}}$, $f(x_i) = \underline{\left(1 + \frac{3i}{n}\right) \tan\left(2 + 1 + \frac{3i}{n}\right)}$

- c). Use your work from Part a) and b) to express $\int_1^4 x \tan(2 + x) dx$ as a limit of Riemann sums with right-endpoints. *Note: You do not need to simplify or evaluate the limit.*

Answer: $\int_1^4 x \tan(2 + x) dx = \underline{\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(1 + \frac{3i}{n}\right) \tan\left(3 + \frac{3i}{n}\right) \cdot \frac{3}{n}}$

3. [6 pts] Consider the function $f(x)$, whose graph is given below. Evaluate the following integrals.



$$1). \int_0^2 f(x) dx = \underline{-3}$$

$$-3 \cdot 2 \cdot \frac{1}{2}$$

$$2). \int_2^9 f(x) dx = \underline{21}$$

$$7 \cdot 6 \cdot \frac{1}{2}$$

$$3). \int_0^{17} f(x) dx = \underline{18 - 8\pi}$$

$$- \frac{\pi \cdot 4^2}{2}$$

4. [6 pts] Given $F(x) = \int_2^{x^2} \frac{\cos t}{t^3} dt$, use Fundamental Theorem of Calculus-Part 1 to find the derivative $F'(x)$.

$$F'(x) = \frac{\cos(x^2)}{(x^2)^3} \cdot 2x = \frac{\cos(x^2)}{x^6} \cdot 2x = \frac{2\cos(x^2)}{x^5}$$

5. [15 pts] 1). Simplify the sum $R_n = \sum_{i=1}^n \left(7 - \frac{16i^3}{n^3}\right) \cdot \frac{2}{n}$. Write your answer as a function of n without any summation signs.

$$\begin{aligned}
 R_n &= \sum_{i=1}^n \left(7 - \frac{16i^3}{n^3}\right) \cdot \frac{2}{n} = \sum_{i=1}^n \left(\frac{14}{n} - \frac{32i^3}{n^4}\right) \\
 &= \left(\sum_{i=1}^n \frac{14}{n}\right) - \left(\sum_{i=1}^n \frac{32i^3}{n^4}\right) \\
 &= \frac{14}{n} \cdot n - \frac{32}{n^4} \left(\sum_{i=1}^n i^3\right) \\
 &= 14 - \frac{32}{n^4} \cdot \frac{n^4 + 2n^3 + n^2}{4} \\
 &= 14 - 8 \left(\frac{n^4 + 2n^3 + n^2}{n^4}\right) \\
 &= 14 - 8 \left(1 + \frac{2}{n} + \frac{1}{n^2}\right) \\
 &= 14 - 8 - \frac{16}{n} - \frac{8}{n^2} \\
 &= 6 - \frac{16}{n} - \frac{8}{n^2}
 \end{aligned}$$

- 2). Evaluate $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(7 - \frac{16i^3}{n^3}\right) \cdot \frac{2}{n}$ by using your result from Part 1).

$$= 14 - 8(1 + 0 + 0) = 6$$

Answer: $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(7 - \frac{16i^3}{n^3}\right) \cdot \frac{2}{n} = \underline{\quad\quad\quad}$.

- 3). Express $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(7 - \frac{16i^3}{n^3}\right) \cdot \frac{2}{n}$ as a definite integral.

$$\Delta x = \frac{2}{n}, \quad a = 0 \quad b = 2 \quad x_i = \frac{2i}{n}$$

$$f(x_i) = f\left(\frac{2i}{n}\right) = 7 - \frac{16i^3}{n^3} = 7 - 2 \cdot \left(\frac{2i}{n}\right)^3 \Rightarrow f(x) = 7 - 2x^3$$

Answer: $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left(7 - \frac{16i^3}{n^3}\right) \cdot \frac{2}{n} = \int_0^2 \underline{(7 - 2x^3)} dx.$

6. Evaluate the following integrals. Make sure to simplify all the trigonometric or inverse trigonometric values in the definite integrals, if any.

1). [8 pts] $\int \frac{x^3 - \sqrt{x} + 1}{x} dx.$

$$= \int \left(x^2 - \frac{1}{\sqrt{x}} + \frac{1}{x} \right) dx$$

$$\left(\int \frac{1}{\sqrt{x}} dx = \int x^{-1/2} dx = 2x^{1/2} + C \right)$$

$$= \frac{1}{3}x^3 - 2x^{1/2} + \ln|x| + C$$

$$= \frac{1}{3}x^3 - 2\sqrt{x} + \ln|x| + C$$

2). [8 pts] $\int_0^{1/2} \left(\frac{6}{\sqrt{1-x^2}} + \frac{1}{2x+1} \right) dx.$

$$= \left[6 \arcsin x + \frac{1}{2} \ln|2x+1| \right] \Big|_0^{1/2}$$

$$= \left[6 \cdot \arcsin\left(\frac{1}{2}\right) + \frac{1}{2} \ln\left|2 \cdot \frac{1}{2} + 1\right| \right] - \left[6 \arcsin(0) + \frac{1}{2} \ln|0+1| \right]$$

$$= 6 \cdot \frac{\pi}{6} + \frac{1}{2} \ln 2 - 6 \arcsin 0 - \frac{1}{2} \ln 1$$

$$= \pi + \frac{1}{2} \ln 2$$

3). [10 pts] $\int x^3 e^{-x^4+5} dx.$ (Hint: Use substitution.)

$$= -\frac{1}{4} \int e^u du$$

$$u = -x^4 + 5$$

$$du = -4x^3 dx$$

$$= -\frac{1}{4} e^u + C$$

$$-\frac{1}{4} du = x^3 dx$$

$$= -\frac{1}{4} e^{-x^4+5} + C$$

u-substitution

4). [10 pts] $\int \frac{\sin x}{1 + \cos^2 x} dx.$

$u = \cos x$

$du = -\sin x dx$

$$= - \int \frac{1}{1+u^2} du$$

$$= - \arctan(u) + C$$

$$= - \arctan(\cos x) + C$$

Integration by parts.

5). [10 pts] $\int x \cos(6x) dx.$

$u = x$

$dv = \cos(6x) dx$

$du = dx$

$v = \frac{1}{6} \sin(6x)$

$$= x \cdot \frac{1}{6} \sin(6x) - \int \frac{1}{6} \sin(6x) dx$$

$$= \frac{x}{6} \sin(6x) + \frac{1}{36} \cos(6x) + C$$

Integration by parts.

6). [10 pts] $\int \frac{\ln x}{x^3} dx.$

$u = \ln x$

$dv = \frac{1}{x^3} dx = x^{-3} dx$

$$= -\frac{1}{2} x^{-2} \ln x - \int \frac{1}{x} \cdot \left(-\frac{1}{2}\right) x^{-2} dx$$

$du = \frac{1}{x} dx$

$v = -\frac{1}{2} x^{-2}$

$$= -\frac{1}{2} x^{-2} \ln x + \frac{1}{2} \int \frac{1}{x^3} dx$$

$$= -\frac{1}{2} x^{-2} \ln x - \frac{1}{2} \cdot \frac{1}{2} \cdot x^{-2} + C$$

$$= -\frac{1}{2} x^{-2} \ln x - \frac{1}{4} x^{-2} + C.$$

$$= -\frac{\ln x}{2x^2} - \frac{1}{4x^2} + C$$

Formula Sheet

$$\sum_{i=1}^n c = cn$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6} = \frac{2n^3 + 3n^2 + n}{6}$$

$$\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2 = \frac{n^4 + 2n^3 + n^2}{4}$$

$$\int \cot x \csc x \, dx = -\csc x + C, \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C, \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C, \quad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C, \quad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$$

$$\int \frac{1}{a^2+x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2-x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined